

CE 181405

SCND

25/6/2025

B.Tech. 4th Semester End-Term Examination

CONSTRUCTION MATERIALS AND CONCRETE TECHNOLOGY

New Regulation (w.e.f. 2017-18) & New Syllabus (w.e.f. 2018-19)

Full Marks – 70

Time – Three hours

The figures in the margin indicate full marks for the questions.

Answer question no. 1 and any four from the rest.

No IS Codes are Allowed in the Examination.

1. Fill in the blanks :

1×10

- (a) If standard consistency of a cement sample is 30%, then the amount of water used for testing of compressive strength cement is _____ %
- (b) Compressive strength percentage developed in concrete within 7 days is approximately _____.
- (c) The two tests used to determine soundness of cement are _____
- (d) The Bogue compound responsible for long term strength is _____
- (e) The negative charge developed around the cement particle after addition of superplasticizer is called _____
- (f) One example of flexible damp proofing materials is _____.
- (g) The test used to find out aggregate abrasion value is _____.
- (h) Third point loading test is used to find out _____ of concrete.
- (i) Tensile yield strength of structural steel is _____.
- (j) Two examples of sound insulating materials are _____.

2. (a) Describe how moisture content and water absorption aggregates affect water cement ratio and strength of concrete.

4

(b) Describe how air entraining agent reduces segregation and bleeding in fresh concrete.

(CO1)

4

(c) How are durability and permeability of concrete related? Discuss the factors affecting the permeability of concrete and how these increase/decrease the durability of concrete.

(CO2)

1+6

(CO2)

[Turn over

3. (a) The test results of sieve analysis for 20 mm single sized aggregate of a 3 kg test sample are given below. 5
(CO3)

Sl.No.	IS Sieve No	Wt. Retained (kg)
1	40 mm	0
2	20 mm	0.30
3	10 mm	2.45
4	4.75 mm	0.23

Determine whether the sample is conforming to IS 383-2016.

- (b) State why fly ash bricks are beneficial for environment. 5
(CO1)
- (c) Discuss the different types of modulus of elasticity of concrete. 5
(CO2)
4. (a) Discuss the requirements of a good workable concrete. How workability of concrete is determined? 3+5
(CO5)
- (b) Mention two mineral admixtures. Discuss the working of mineral admixtures. 2+5
(CO5)
5. (a) Explain with example what is destructive and non-destructive testing of concrete? Under what circumstances non-destructive testing of concrete is to be adopted? 4+4
(CO3)
- (b) Draw the section of an exogenous tree and discuss different parts of it. 3+4
(CO1)
6. The following data are given for a concrete mix design : 15
(CO4)
- (a) Characteristic compressive strength required at the field after 28 days of curing = 30 N/mm^2 .
- (b) Type of aggregate = Maximum nominal size 20mm angular
- (c) Standard deviation to be used = 5.00
- (d) Probability factor (k) = 1.65
- (e) Value of X = 6.5
- (f) Approximate air content = 1%
- (g) Workability required = 150 mm of slump
- (h) Exposure condition = Moderate
- (i) Specific gravities of cement, coarse aggregate, fine aggregate and superplasticizer are 2.95, 2.64, 2.59 and 1.18 respectively.

- (j) Grading zone for fine aggregate = Zone III
- (k) Maximum water content per cubic meter of concrete for 20mm aggregate is 186 kg
- (l) Volume of CA per unit volume of total aggregate, for FA zone III and nominal maximum size of 20mm aggregate is 0.64.
- (m) Minimum cement content from durability consideration = 300 kg
- (n) Maximum cement content = 450 kg
- (o) Method of concrete placement = Pumping

Determine the actual quantities required for the mix at saturated surface dry (SSD) condition per cubic meter of concrete for W/C ratio of 0.49.

7. Write short notes (Any *three*) : 3×5
- (a) AAC blocks (CO1)
 - (b) High density concrete (CO5)
 - (c) Shrinkage of concrete (CO2)
 - (d) Corrosion of reinforcement. (CO2)